

LEAF, CONFIGURATION AND SYSTEM

CI head f028ec5 / merge ae11d8b | Verilator 5.020 | Group 3/3

21 RTL benches passed; independent arithmetic audit passed
RTL CI job succeeded; workflow conclusion is separate

BENCH / OBSERVED LOG COUNTS

REFERENCE CHECK / COVERAGE BOUNDARY

tree_mapping_tb

12,904 stimulus rows

Observed: TREE_MAPPING_COMPLETE checks=12904

Loop popcount and direct cell sums;
T/G/R + invalid-ID comparison.

This archive is RTL, not mapped-netlist proof.

review_leaf_tb

91,142 check calls

Observed: REVIEW_LEAF_COMPLETE

Popcount/thermometer rules, address
bounds, allocation wrap and floor oracle.

The logged count is assertion calls.

review_dither_tb

7 amplitude/seed conditions
accepted/condition: 196,866 or 200,000

Observed: REVIEW_DITHER_COMPLETE

Support/PMF/mean/lag-1 checks;
hold and seed-reset invariants.

Accepted draws differ from 200k clock cycles.

p1_tb

464,823 TB checks; 0 errors

Observed: P1 RESULT: PASS

M1-M5 golden vectors plus structural
properties, dither support and connected chain.

464,823 is the TB check counter, not codes.

p2_periph_tb

709 TB checks; 0 errors

Observed: p2_periph PASS

Directed guards, sticky status, RDAC
fanout, phase schedule and register rules.

Full-table load cannot trigger unreachable sum guard.

p2_smoke_tb

1,154 TB checks; 0 errors

Observed: p2_smoke PASS

Configuration readback; allocator
invariants and selected popcount inputs.

Smoke coverage, not complete acceptance.

p3_top_tb

16,386 TB checks; 0 errors
4,095 output rows; 16-cycle spacing

Observed: P3 TOP RESULT: PASS

File oracle for dout/clip + cumulative
OR for sticky analog-overflow status.

Forced dither; file row 0 skipped; no dither-source test.

Counts retain their own units. No combined 'total coverage' score. All 21 log and TB hashes verified.

Source: docs/evidence/20260930/ci_rtl_f028/ | RTL only; not STA or analog performance.